CE 6018 Seismic Data Analysis, Tutorial 3

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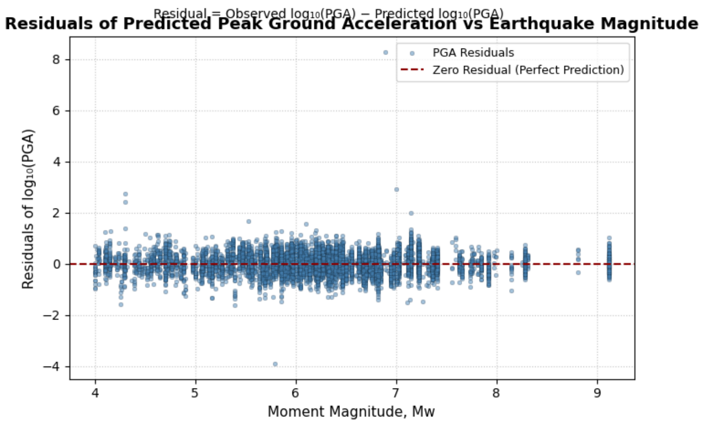
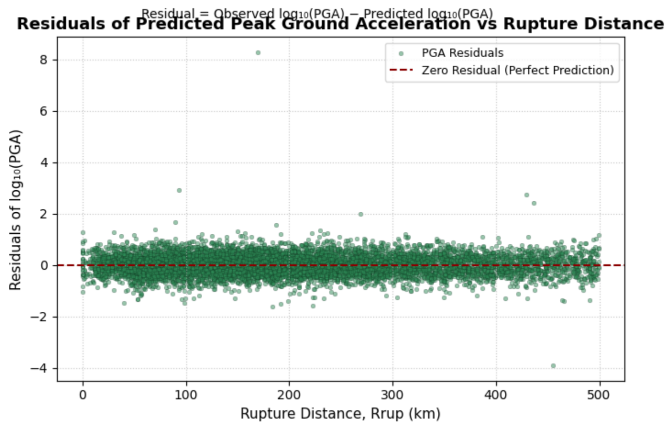
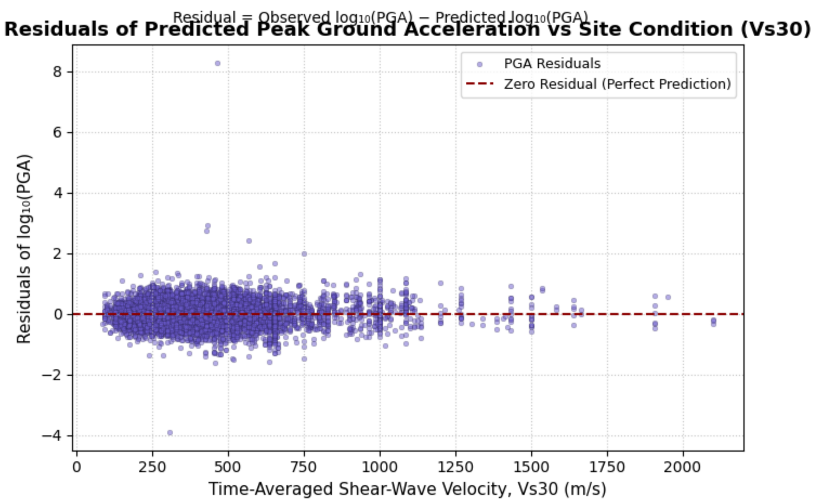
1) Complete the ANN coding and generate all the graphs as mentioned.

#input variables
EaEarthquake_Magnitude - Mw
Vs30_Selected_for_Analysis_m_s -> Log_Vs30
Rjb_km -> R_Rup
Rjb_km -> Log_R_Rup
Fault_Type -> F
#these all are the input variables
#Output variable
PGA_g, PGV_cm_sec, and 20 cols as described below
#T0pt010S - this means T = 0.010s, so we need to take 20 cols starting from T0pt010S,..
Performance :
Epoch 100/100
115/115 ━━━━━━━━━━━━━━━━━━━━ **0s** 2ms/step - loss: 0.1931 - mae: 0.3109 - val_loss: 0.1718 - val_mae: 0.3072
Test MSE: 0.16276204586029053
Test MAE: 0.30786100029945374
Overall R^2: 0.6726312579806052
Residuals

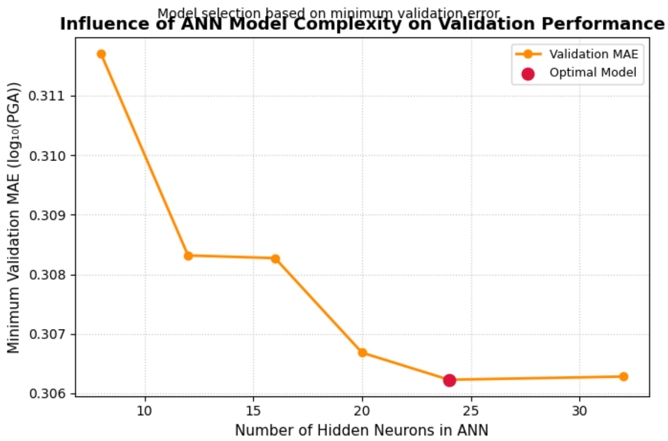


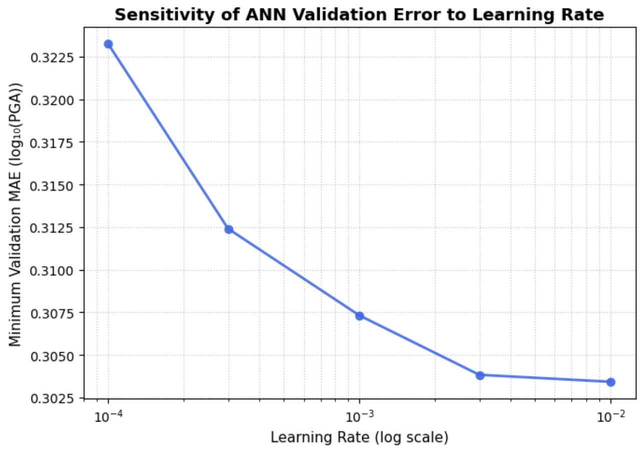


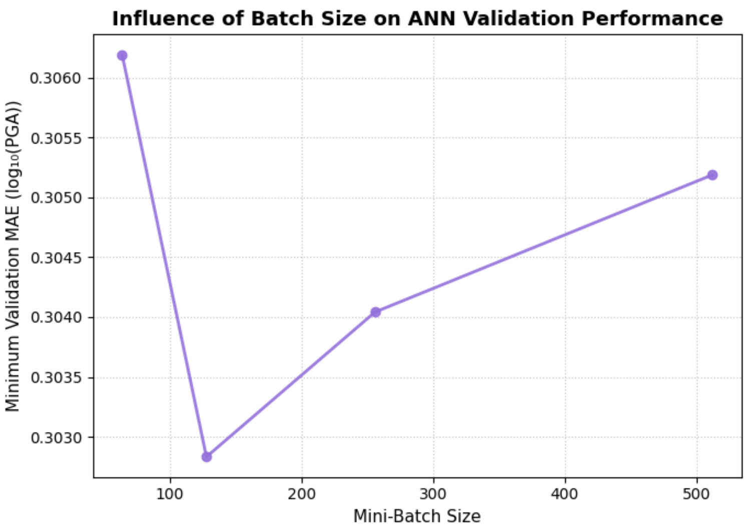


2) Hyper parameter tuning

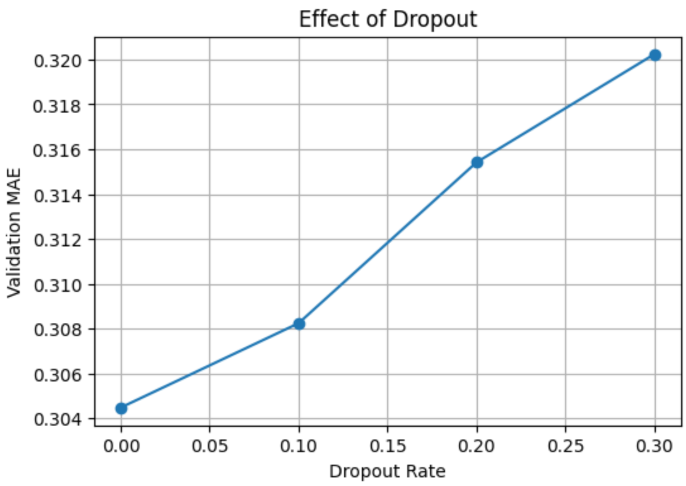
B. Network hyperparameters
Batch size
Learning rate
Number of hidden neurons



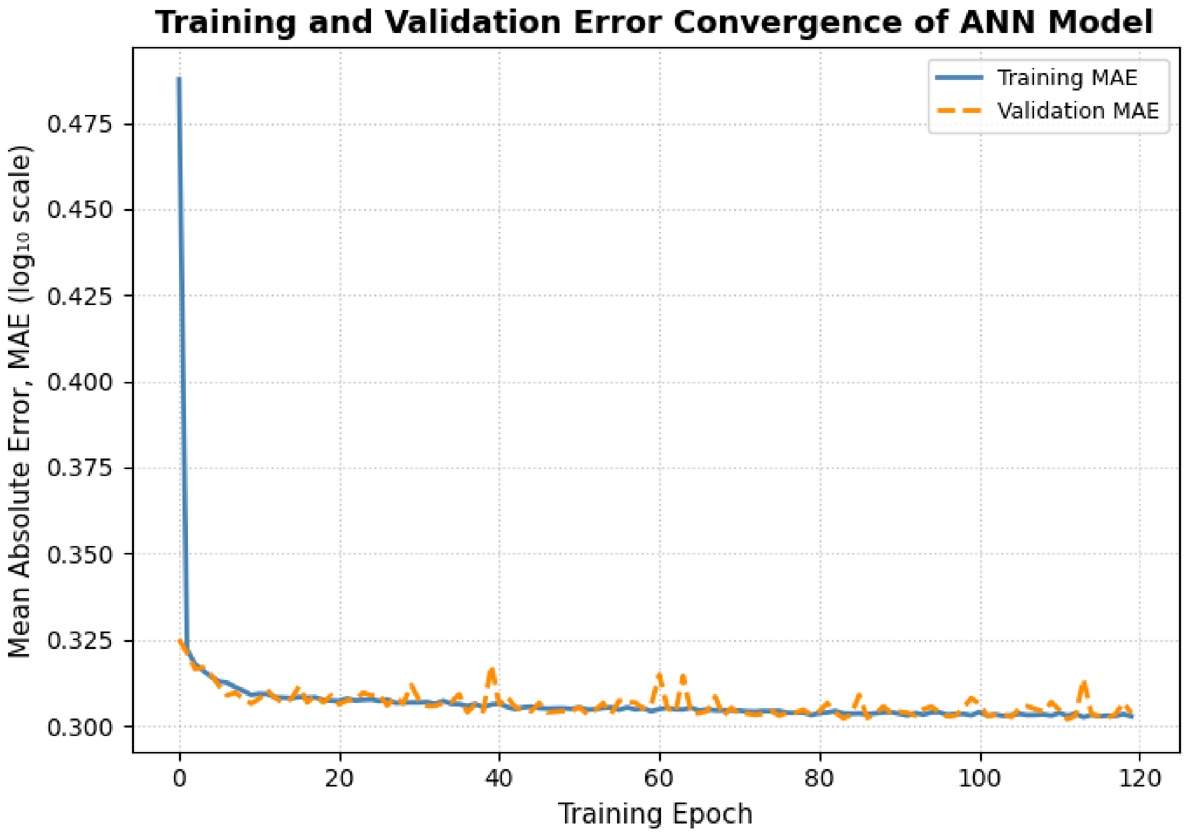




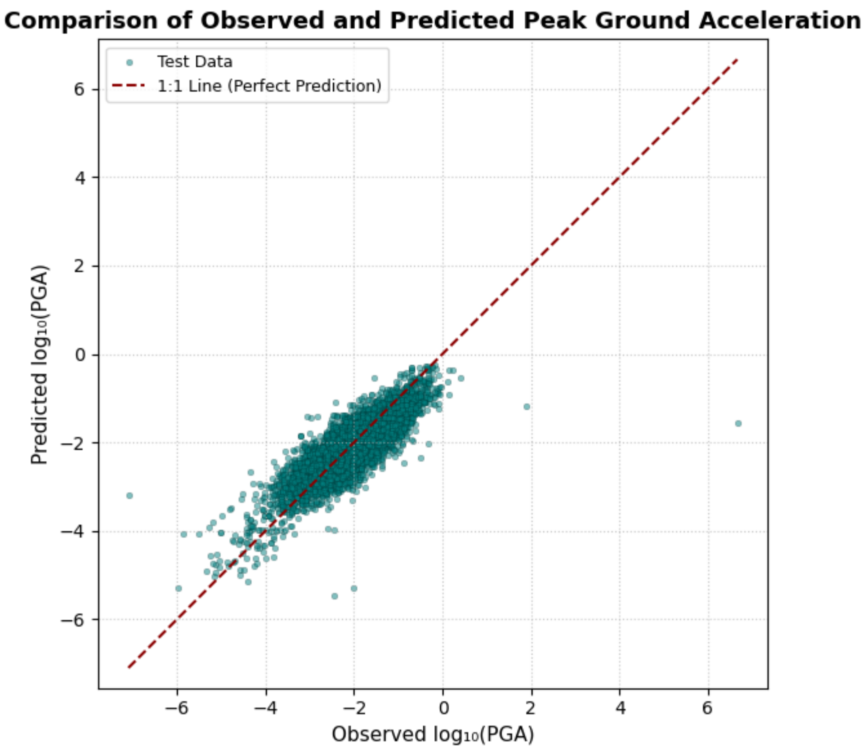
dropout_list = [0.0, 0.1, 0.2, 0.3]

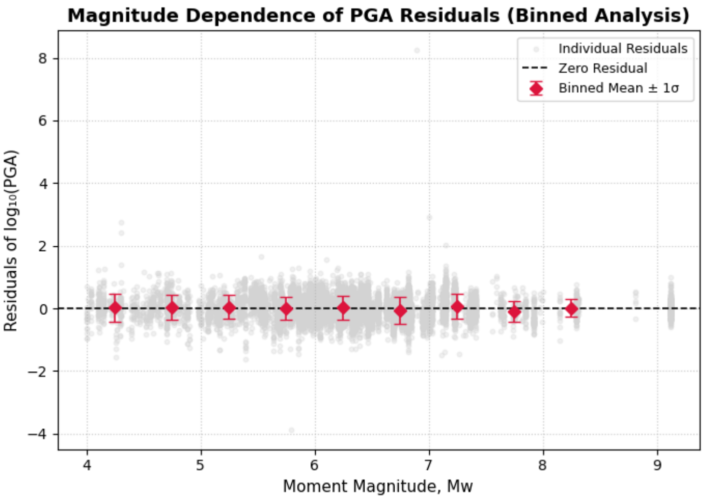


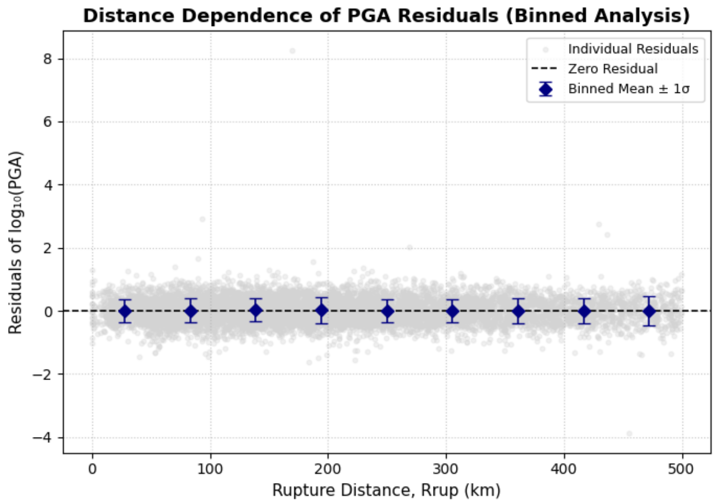
Final Model with best parameters after hyperparameter tuned

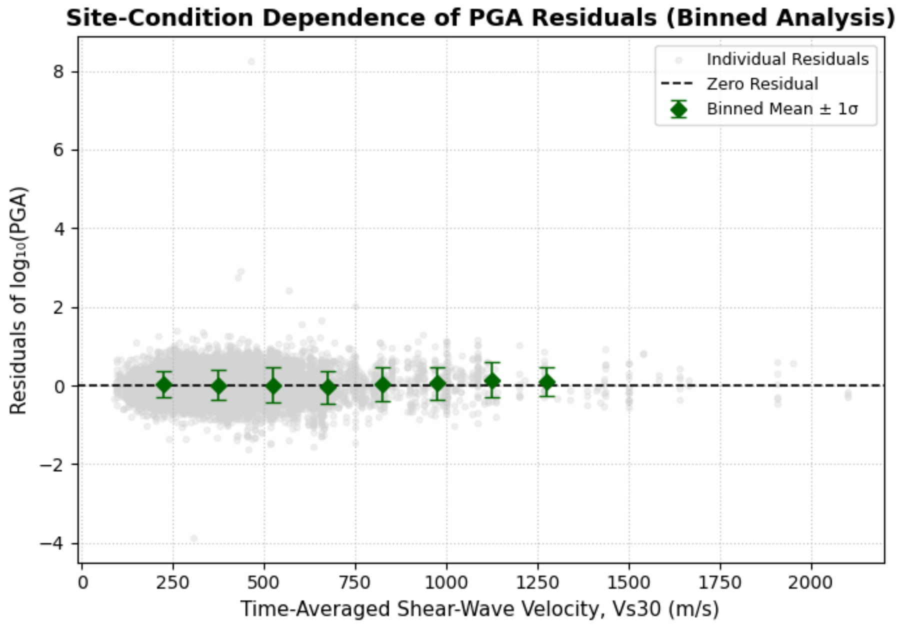


Final Test MSE: 0.16064026951789856
Final Test MAE: 0.30466750264167786

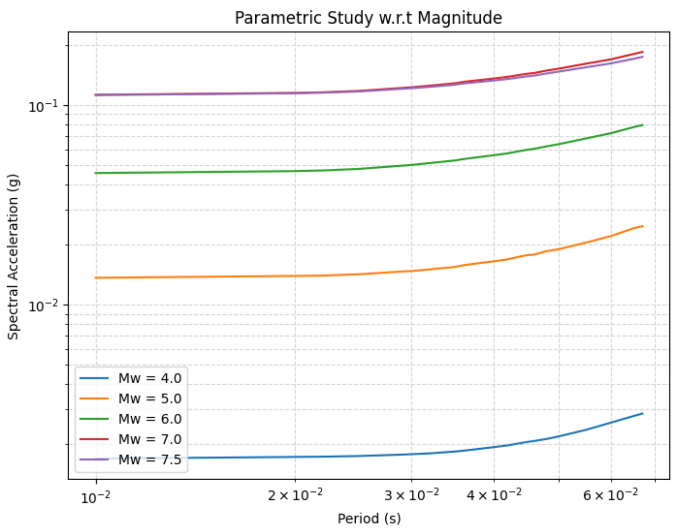


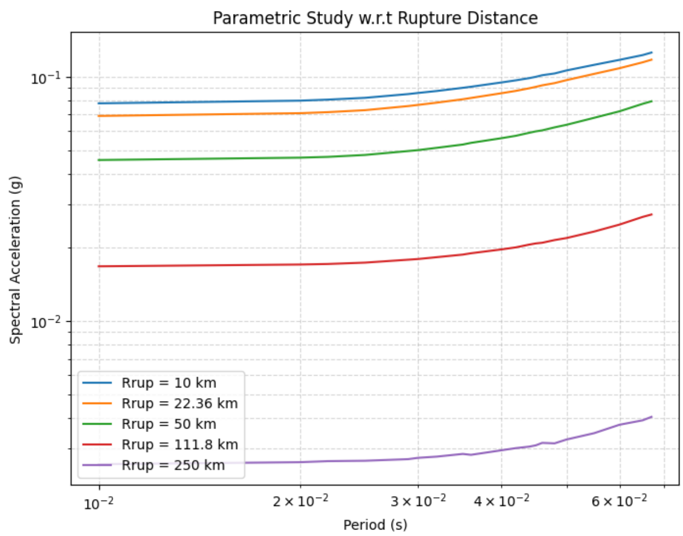


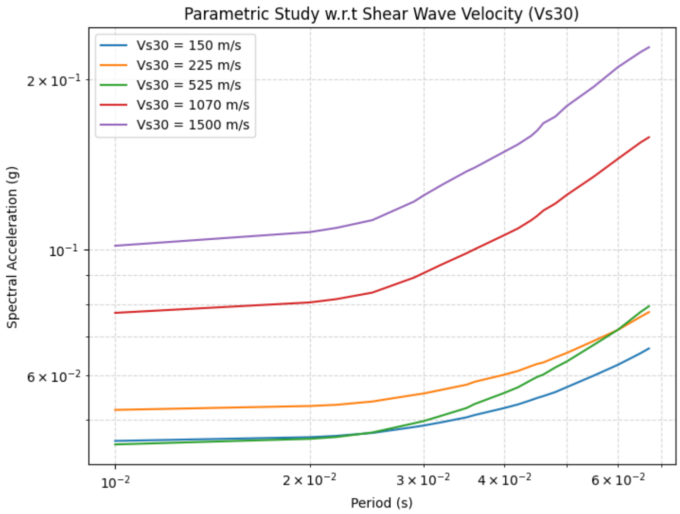


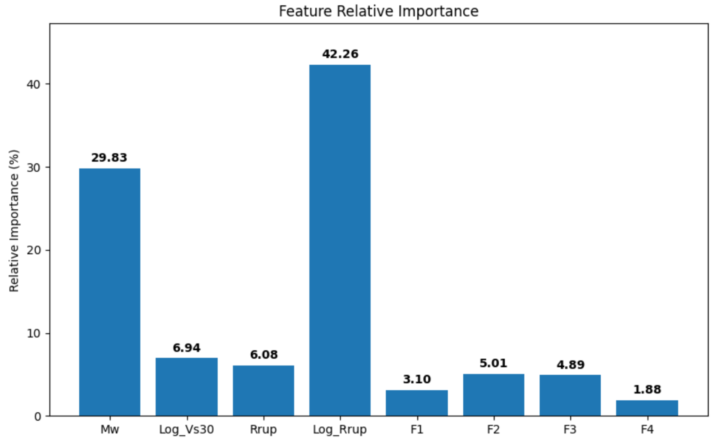


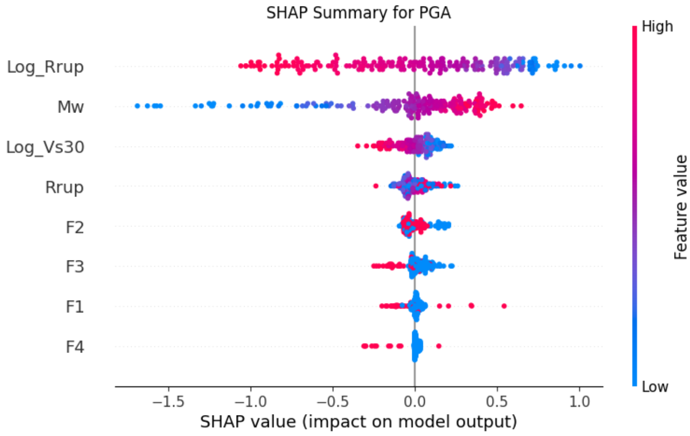
PARAMETRIC STUDIES

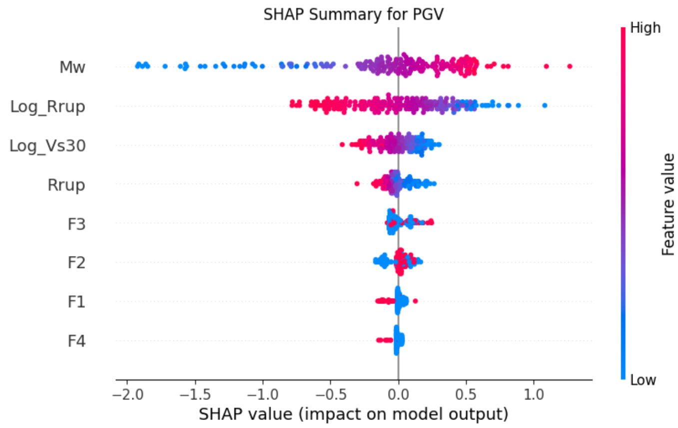


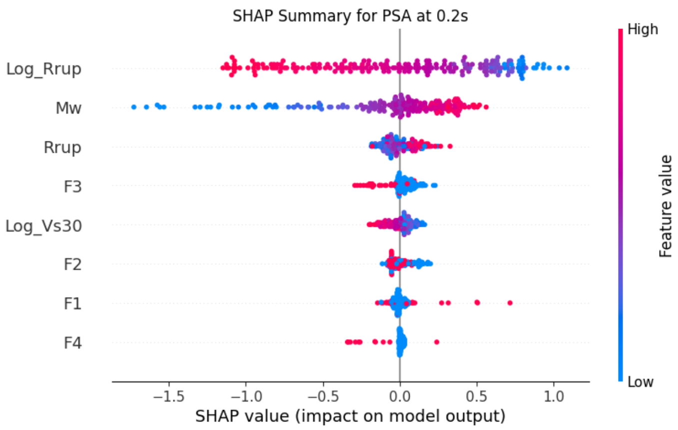


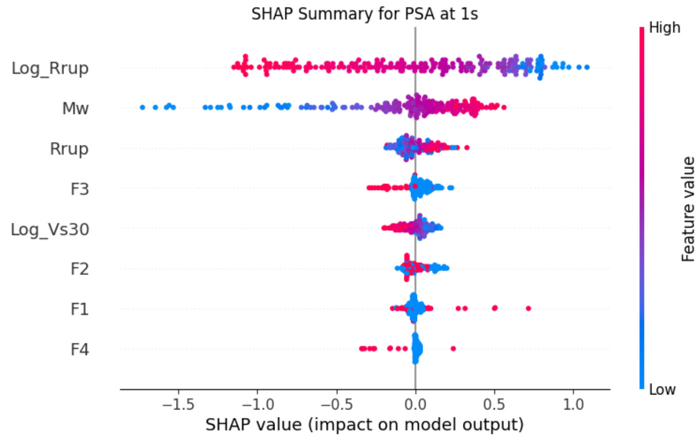

Parametric Study w.r.t Shear Wave Velocity (Vs30)
Vs30 = 150 m/s
Vs30 = 225 m/s
Vs30 = 525 m/s
Vs30 = 1070 m/s
Vs30 = 1500 m/s
Spectral Acceleration (g)
Period (s)


Feature Relative Importance
Relative Importance (%)
29.83
6.94
6.08
42.26
3.10
5.01
4.89
1.88
Mw
Log_Vs30
Rrup
Log_Rrup
F1
F2
F3
F4


SHAP Summary for PGA
Log_Rrup
Mw
Log_Vs30
Rrup
F2
F3
F1
F4
SHAP value (impact on model output)
High
Feature value
Low

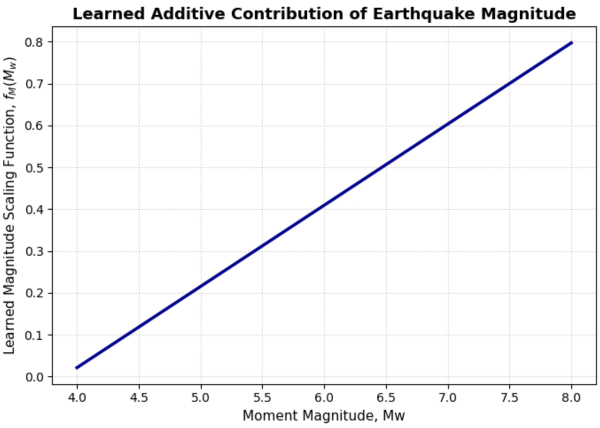

SHAP Summary for PGV
Mw
Log_Rrup
Log_Vs30
Rrup
F3
F2
F1
F4
SHAP value (impact on model output)
High
Feature value
Low

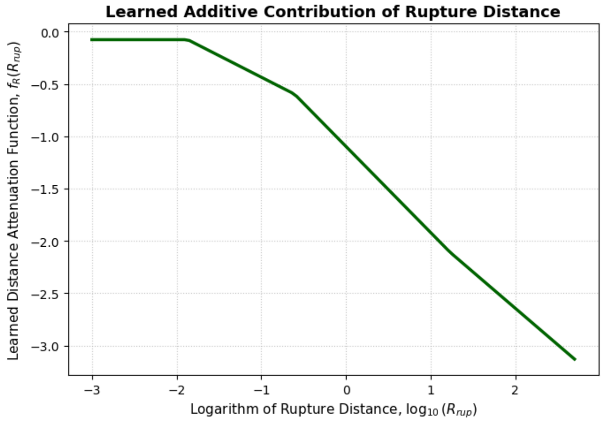

SHAP Summary for PSA at 0.2s
Log_Rrup
Mw
Rrup
F3
Log_Vs30
F2
F1
F4
SHAP value (impact on model output)
High
Feature value
Low

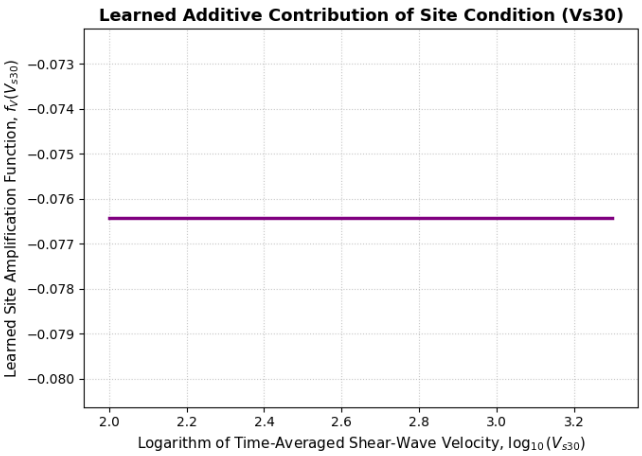

SHAP Summary for PSA at 1s
Log_Rrup
Mw
Rrup
F3
Log_Vs30
F2
F1
F4
SHAP value (impact on model output)
High
Feature value
Low

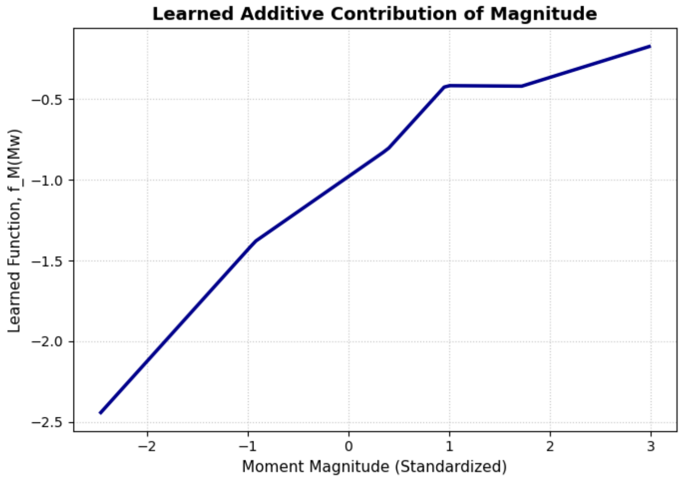
2) Additive Neural Network

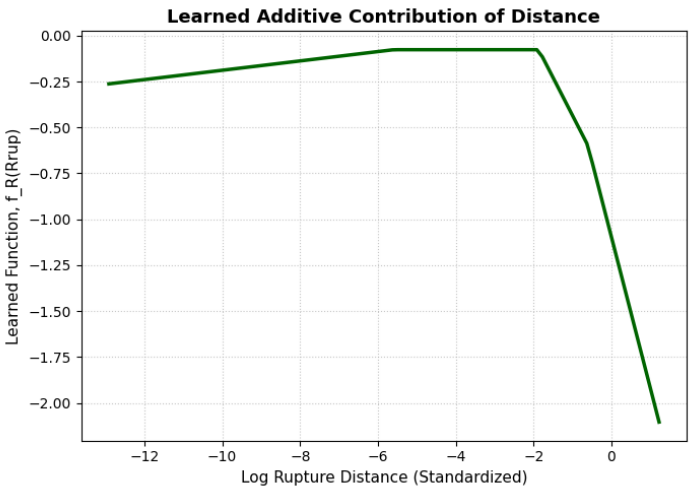
Additive NN Test MAE (PGA): 0.3101617395877838

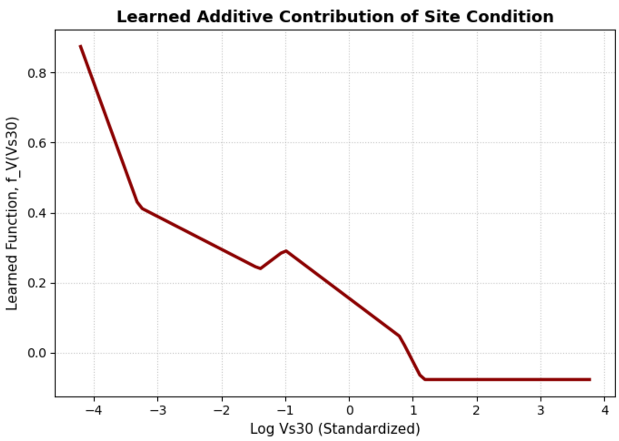












3) Here are my observations from the plots of ANN, additive neural network

1. Model Performance & Residuals

- **Plot: "Comparison of Observed and Predicted Peak Ground Acceleration"**
  - **Observation:** The model's predictions are highly accurate. The data points cluster tightly along the 1:1 dashed line, meaning the predicted PGA matches the actual observed PGA very closely across small and large values.
- **Plots: "Residuals vs Magnitude", "Residuals vs Distance", and "Residuals vs Vs30"**
  - **Observation:** The errors (residuals) are evenly scattered above and below the zero line. Because the binned means stay flat at zero and don't slope up or down, it proves the model is unbiased. It does not systematically over-predict or under-predict for specific earthquake sizes, distances, or soil types.

2. Hyperparameter Tuning

- **Plots: "Effect of Hidden Neurons", "Effect of Learning Rate", "Effect of Batch Size", "Effect of Dropout"**
  - **Observation:** The tuning graphs show exactly where the model finds its "sweet spot." The error drops to its lowest point around 25 hidden neurons. Interestingly, the "Effect of Dropout" plot shows a straight line going up; this means adding dropout actually increases the error, so the network performs best with zero dropout.

3. Parametric Studies (The Physics of the Model)

- **Plot: "Parametric Study w.r.t Magnitude"**
  - **Observation:** The model successfully learned that larger earthquakes cause stronger shaking. As the moment magnitude (M_w) goes up from 4.0 to 7.5, the spectral acceleration curves shift uniformly higher across all time periods.
- **Plot: "Parametric Study w.r.t Rupture Distance"**
  - **Observation:** The model perfectly captures seismic wave attenuation. As the distance from the rupture increases from 10 km to 250 km, the ground shaking drops significantly.
- **Plot: "Parametric Study w.r.t Shear Wave Velocity (Vs30)"**
  - **Observation:** The model is highly sensitive to site conditions (soil stiffness). The varying curves show that different V_s30 values drastically change the spectral acceleration, proving the network isn't just relying on distance and magnitude, but also on local soil conditions.

4. Feature Importance & SHAP Analysis

- **Plot: "Feature Relative Importance" (Bar Chart)**
  - **Observation:** The model relies heavily on just two major factors: the logarithm of distance (Log_Rrup) at **42.26%** and Earthquake Magnitude (Mw) at **29.63%**. Soil condition (Log_Vs30) and fault types act as minor tweaks rather than main drivers.
- **Plots: "SHAP Summary for PGA, PGV, PSA"**
  - **Observation:** These plots show *how* the features influence the prediction. For Magnitude (Mw), the red dots (high values) are on the right side, meaning a big earthquake pushes the predicted shaking higher. For Distance (Log_Rrup), the red dots are on the left side, meaning being far away pushes the predicted shaking lower. This exactly matches real-world earthquake physics.

5. Additive Neural Network Contributions

- **Plot: "Learned Additive Contribution of Earthquake Magnitude"**

    - **Observation:** By isolating the magnitude branch, we see a nearly perfect straight line sloping upward. This means the model learned a direct, linear relationship: adding more magnitude directly adds to the ground motion prediction.
- **Plot: "Learned Additive Contribution of Rupture Distance"**
    - **Observation:** Isolating the distance branch reveals a downward curve. As you move further right on the x-axis (greater distance), the learned function drops heavily, which visualizes the exact attenuation penalty the network applies to far-away sites.